
NeRo-CLIP: Negation-Routed adapter for frozen CLIP

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Abstract

CLIP misranks captions that contain negation operators (“no”, “not”, “without”). On the MCQ-Neg benchmark from NegBench, a 4-way image–caption matching benchmark, CLIP ViT-B/32 scores 39.66, and two of three published negation-aware fine-tuned variants score 25.07 and 26.40, below CLIP. Existing corrections either fine-tune the full backbone with negation-aware objectives on $\sim 566\text{K}$ – 12M image–caption pairs or steer CLIP’s text-encoder activations along a linear negation direction at inference. Both approaches intervene broadly: fine-tuning modifies the entire backbone, and the published steering recipe shifts every text activation, which under a common protocol drops general COCO retrieval by 4.30pp R@1. Rather than treating negation correction as a global adaptation of CLIP, we cast it as a selective intervention problem for mixed retrieval traffic: negated captions are corrected while ordinary captions stay on CLIP’s original embedding. We propose **NeRo-CLIP**, a routed residual negation correction for frozen CLIP: a rank-8 residual adapter (8K parameters) at the final-EOS output, applied only when an 11-marker regex router detects negation operators in the caption. We train it under a joint loss that mixes MCQ with image retrieval. NeRo-CLIP improves over CLIP on all four negation benchmarks we evaluate (MCQ-Neg, Retrieval-Neg R@1, SimpleNeg Top-1, N-COCO R@1). Per-caption gating leaves CLIP’s original embedding on 98.96% of COCO captions; NeRo-CLIP holds general COCO val2017 retrieval at 30.37 R@1 versus CLIP’s 30.36.

1 Introduction

CLIP Radford et al. [2021] aligns images and text in a shared embedding space, learned by contrastive pretraining on web-scale image-caption pairs. It is now a standard foundation for cross-modal retrieval, text-to-image generation, and zero-shot classification. However, this shared space treats text largely as an unordered bag of concepts: a content word in the query lifts the score of images containing that concept, whether the sentence asserts or negates it Koishigarina et al. [2025], Yuksekgonul et al. [2023]. CLIP also fails on *half-truths*—captions extended with a plausible but wrong detail, where similarity paradoxically rises instead of dropping Kargi et al. [2026]. Asked to retrieve “a beach with no people”, CLIP returns the most beach-like image regardless of crowd density.

On MCQ-Neg, a 4-way image-caption matching task from NegBench Alhamoud et al. [2025] built on MS-COCO val2017, CLIP ViT-B/32 scores 39.66. Although this is above the 25% random baseline,

it leaves substantial room for improvement on negation-sensitive image–text matching. This failure is more pronounced on the negation template: two of the published negation-aware fine-tuned variants score 25.07 and 26.40 on the same test split, below the CLIP baseline. This behavior has well-documented structural causes: negation markers appear in fewer than 1% of web-scale image-caption pairs Aggarwal et al. [2026], and the contrastive objective rewards reading text as unordered concept bags Koishigarina et al. [2025], Yuksekgonul et al. [2023].

Prior work has addressed this failure along two main directions. The first fine-tunes CLIP with negation-aware data: NegCLIP Yuksekgonul et al. [2023] on $\sim 566\text{K}$ pairs, ConCLIP Singh et al. [2025] on 228K, and CLIP-CC12M with a negation objective Alhamoud et al. [2025] on 12M. These methods expose a trade-off between MCQ-style discrimination and retrieval alignment rather than providing uniform improvements across settings. NegCLIP excels at retrieval (60.77 Retrieval-Neg R@1) but collapses on MCQ-Neg to 26.40 on the paper test split ($n=1,775$); ConCLIP shows the same pattern at 25.07. Only CLIP-CC12M lifts MCQ-Neg above CLIP, reaching 55.13, but requires 12M negation-aware training pairs and full-backbone updates. NeRo-CLIP reaches 54.70 on the same metric with 8K parameters and $\sim 24\text{K}$ training pairs (4K MCQ rows + 20K retrieval pairs) on a frozen backbone.

The second line is test-time activation steering. A recent method Sammani et al. [2026] fits a logistic regression at each of the 12 text-encoder layers to identify the direction along which negation varies, then shifts the [EOS] activation along that direction at every layer with a fixed strength. It requires no heavy backbone fine-tuning and uses about 4,000 training pairs. The method scores 41.60 on MCQ-Neg and remains particularly strong on open-pool negation retrieval benchmarks (SimpleNeg, N-COCO).

A negation correction, however, should not be assessed solely by its gain on explicitly negated captions. In realistic retrieval traffic, negated queries are rare and affirmative captions dominate, so a method that corrects negation by displacing every text representation may improve rare negated queries while degrading the much more common affirmative queries. We therefore formulate negation correction as a *selective post-hoc intervention* problem: the correction should act when a query carries explicit negation and otherwise preserve CLIP’s original embedding geometry. From this perspective, a useful correction must satisfy two requirements: accuracy on negated captions and selectivity on affirmative ones. Our results support this framing (Table 2): routing confines the intervention to a small fraction of captions, holding general retrieval at the CLIP baseline where unconditional steering does not.

We propose **NeRo-CLIP**, a rank-8 residual adapter (8,192 parameters at $d=512$) inserted after the final-EOS output of frozen CLIP. A regex over 11 negation markers decides per caption whether to apply it. When the router detects a negation marker, the adapter rewrites the text embedding from z to $g_\theta(z)$. Otherwise the original CLIP text embedding is left unchanged.

We train NeRo-CLIP with a joint objective

$$L = \lambda L_{\text{MCQ}} + (1-\lambda) L_{\text{retrieval}},$$

where L_{MCQ} provides contrastive supervision over MCQ-Neg candidate captions and $L_{\text{retrieval}}$ is an in-batch image–text InfoNCE loss on Retrieval-Neg pairs. We sweep $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$, where λ controls the relative weight of the two objectives. Relative to MCQ-only training ($\lambda=1$), the mixed objective at $\lambda=0.75$ sacrifices a small amount of MCQ-Neg accuracy while substantially improving retrieval:

- MCQ-Neg: 56.85 $\rightarrow$ 54.70 (−2.15pp).
- Retrieval-Neg R@1: 25.68 $\rightarrow$ 50.08 (+24.40pp).
- SimpleNeg Top-1: 44.4 $\rightarrow$ 49.7 (+5.3pp).
- N-COCO R@1: 0.50 $\rightarrow$ 0.60 (+0.10).

Routed deployment also leaves general COCO val2017 retrieval intact: NeRo-CLIP scores 30.37 R@1 versus CLIP’s 30.36 (+0.01pp), while layerwise activation steering, the strongest open-pool baseline, drops 4.30pp on the same protocol. NeRo-CLIP improves over CLIP on all four negation benchmarks while preserving general COCO retrieval (Tables 1 and 2).

Contributions.

1. **A selective-intervention formulation of negation correction.** We frame CLIP negation correction as a mixed-traffic retrieval problem: a correction should improve explicitly negated queries without degrading ordinary affirmative retrieval. This shifts the goal from global model adaptation to selective intervention. Empirically, routing preserves COCO retrieval at the CLIP baseline (30.37 vs. 30.36 R@1), whereas unconditional steering reduces it by 4.30pp (Table 2).
2. **A routed residual adapter for frozen CLIP.** We instantiate this formulation with a rank-8 residual adapter (8K parameters) applied at the final-EOS text embedding of CLIP and activated per caption by an 11-marker explicit-negation regex router. The CLIP image and text encoders remain frozen end-to-end.
3. **A joint MCQ–retrieval training objective.** We train the adapter with a convex mixture of an MCQ contrastive loss and an in-batch retrieval InfoNCE loss, and sweep a single mixing coefficient λ to balance within-row MCQ-Neg discrimination against Retrieval-Neg image–text alignment. At $\lambda=0.75$, NeRo-CLIP recovers the retrieval signal damaged by MCQ-only training (+24.40pp on Retrieval-Neg R@1 vs. MCQ-only) at the cost of −2.15pp on MCQ-Neg.

2 Related Work

Negation benchmarks. Existing negation benchmarks probe different aspects of CLIP’s negation failure, and we use four of them in our evaluation. Alhamoud et al. [2025] introduce **NegBench**, a benchmark suite built on MS-COCO val2017 Lin et al. [2014] and Pascal VOC2007 Everingham et al. [2015] with two core tasks: MCQ-Neg (4-way image-caption matching with positive, hybrid, and negative caption templates) and Retrieval-Neg (text-to-image retrieval with negated captions). They document that multiple CLIP variants score near or below random on the negation template—a failure mode they call *affirmation bias*: the model treats every caption as if its content words were asserted, regardless of negation markers. Sammani et al. [2026] contribute two additional open-pool retrieval benchmarks: **SimpleNeg** (900 negated queries against a 25K-image pool, evaluated by an MLLM judge) and **N-COCO** (10 distribution-shift queries with R@1 measured directly). NegBench evaluates discrimination within a fixed candidate set; SimpleNeg and N-COCO evaluate retrieval from a large open pool.

Prior negation corrections. Prior work has mostly addressed CLIP’s negation failures by either changing the model through additional training or steering its text representations at inference. NegCLIP Yuksekgonul et al. [2023] fine-tunes CLIP ViT-B/32 on $\sim 566\text{K}$ COCO caption pairs with composition-aware hard negatives, ConCLIP Singh et al. [2025] adds a hard-negative contrastive loss on 228K pairs, and CLIP-CC12M Alhamoud et al. [2025] scales the idea to 12M pairs with a negation objective. Since these methods update the backbone, the resulting model is used for both negated and ordinary affirmative captions. This can improve some negation settings, but it does not guarantee a uniform improvement: Sammani et al. [2026] report that NegCLIP and ConCLIP regress on MCQ-Neg relative to CLIP, and concurrent work Aggarwal et al. [2026] connects this behavior to forgetting of CLIP’s original image–text alignment. A second route avoids retraining the backbone and instead changes the text representation at inference. Sammani et al. [2026] fit a logistic-regression probe at each of CLIP’s 12 text-encoder layers using $\sim 4\text{K}$ positive/negative caption pairs, then shift the [EOS] activation along the learned negation direction with strength $\alpha=0.13$. This is analogous to steering-vector interventions in language models Li et al. [2023], Panickssery et al. [2023]. In the released configuration we evaluate, the shift is applied to every text input across all text-encoder layers. We therefore measure its COCO retrieval retention in Section 4.3, since a correction that helps negated captions can still be undesirable if it degrades ordinary retrieval.

Parameter-efficient adaptation of CLIP. Adapter methods inject small learnable modules into a frozen backbone. CLIP-Adapter Gao et al. [2024] and LoRA-style variants Hu et al. [2022] tune CLIP for downstream classification with a few thousand parameters, applied uniformly to every input. **NeRo-CLIP** sits in this family but adds two restrictions: the adapter occupies a single location—the final-EOS output—rather than every layer, and a per-input regex router decides whether to apply it, so captions without explicit negation markers pass through CLIP unchanged.

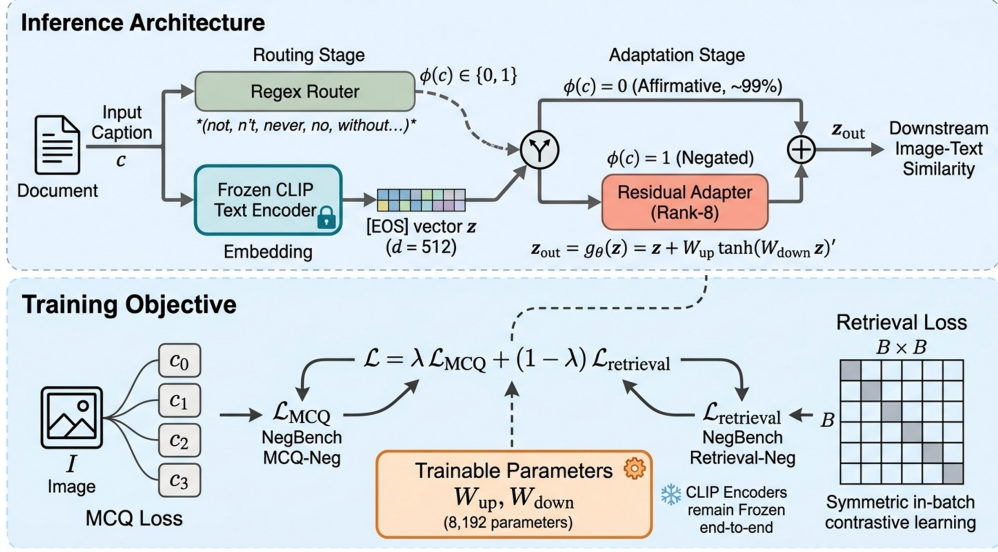


Figure 1: **NeRo-CLIP architecture.** The frozen CLIP text encoder produces a d -dimensional [EOS] embedding z . A rank-8 residual adapter $g_\theta(z) = z + W_{\text{up}} \tanh(W_{\text{down}} z)$ produces a text-side correction. The correction is applied only when the regex router detects an explicit negation marker; otherwise the original CLIP embedding is left unchanged. The adapter is trained with a joint MCQ–retrieval objective controlled by λ .

3 Method

NeRo-CLIP has three components: a residual adapter at the final-EOS text embedding of a frozen CLIP text encoder (Section 3.1), a regex router that selects captions containing explicit negation (Section 3.2), and a joint training objective combining MCQ contrastive supervision with in-batch retrieval InfoNCE (Section 3.3). Figure 1 gives an overview of the architecture and training objective.

3.1 Adapter

Let f denote the frozen CLIP text encoder. It maps a tokenized caption c to its final-EOS embedding $z = f(c) \in \mathbb{R}^d$ ($d=512$ for ViT-B/32). We define a rank- r residual adapter at this point as

$$g_\theta(z) = z + W_{\text{up}} \tanh(W_{\text{down}} z), \quad W_{\text{down}} \in \mathbb{R}^{r \times d}, \quad W_{\text{up}} \in \mathbb{R}^{d \times r}.$$

With $r=8$ and $d=512$, the adapter contains $2rd = 8,192$ trainable parameters. CLIP’s text encoder weights stay frozen; only W_{down} and W_{up} are updated. We initialize W_{up} to zero, so g_θ starts as the identity.

At inference, routed captions pass through the adapter:

$$z_{\text{out}} = g_\theta(z) = z + W_{\text{up}} \tanh(W_{\text{down}} z).$$

Unrouted captions keep their original embedding z unchanged.

3.2 Regex router

NeRo-CLIP applies the adapter only to captions with explicit surface negation. We use a conservative case-insensitive word-boundary regex over 11 markers:

$$\{not, n't, never, neither, no, none, nothing, empty, without, absent, missing\}.$$

These markers cover the canonical sentential, existential, and privative forms used in NegBench’s negation templates and in the negated captions of Retrieval-Neg.

Formally, the router is

$$\phi(c) = \mathbb{I}[c \text{ contains any of the 11 markers}].$$

On COCO val2017, the router matches 260 out of 25,000 captions (1.04%), leaving the remaining 98.96% unchanged. On the NegBench MCQ-Neg negation-template rows, the router matches the explicitly negated captions. This high-precision rule intentionally misses paraphrastic forms such as *devoid of* or *lacking*; we return to this limitation in Section 5.

3.3 Joint training

The adapter must satisfy two related but different objectives: separating the correct caption from its three distractors within an MCQ-Neg row, and ranking the correct image highly against a batch of distractor images. We combine the two through a convex mixture

$$L = \lambda L_{\text{MCQ}} + (1-\lambda) L_{\text{retrieval}}, \quad \lambda \in [0, 1].$$

$\lambda=1$ recovers the MCQ-only baseline; $\lambda=0$ recovers the retrieval-only baseline.

MCQ loss. Each MCQ-Neg row consists of an image I , four candidate captions c_0, c_1, c_2, c_3 , and an index $y \in \{0, 1, 2, 3\}$ marking the correct caption. We define

$$L_{\text{MCQ}} = -\log \frac{\exp(\tau \cdot \cos(z_I, g_\theta(z_{c_y})))}{\sum_{k=0}^3 \exp(\tau \cdot \cos(z_I, g_\theta(z_{c_k})))}.$$

Here z_I is the frozen CLIP image embedding and $z_{c_k} = f(c_k)$ is the frozen text embedding of the k -th candidate.

Retrieval loss. For a batch of B image–caption pairs $\{(I_i, c_i)\}_{i=1}^B$ from the NegBench Retrieval-Neg split, we use the standard symmetric in-batch InfoNCE objective:

$$L_{\text{retrieval}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{e^{\tau s_{ii}}}{\sum_j e^{\tau s_{ij}}} + \log \frac{e^{\tau s_{ii}}}{\sum_j e^{\tau s_{ji}}} \right], \quad s_{ij} = \cos(z_{I_i}, g_\theta(z_{c_j})).$$

Role of λ . The coefficient λ controls the training balance between within-row MCQ discrimination and cross-image retrieval alignment. We sweep $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ and report the effect of this trade-off in Section 4.2.

3.4 Design choices

NeRo-CLIP is designed for explicit negation correction in a frozen CLIP retrieval pipeline while preserving general retrieval behavior.

Text-side intervention. Explicit negation is introduced by lexical operators (“no”, “not”, “without”); the corresponding visual evidence is the absence of an object or relation, with no directly observable visual counterpart. Modifying the image encoder would shift representations for all images, including those queried with affirmative captions. A text-side correction localizes the intervention to the representation where the negation operator appears.

Final-EOS placement. We place the adapter at the final-EOS embedding rather than inside transformer layers. The [EOS] embedding is the representation that directly enters the image–text similarity score, so changing it gives a direct handle on retrieval ranking while leaving the frozen text encoder intact. A single-location adapter also keeps routing simple: the router either applies one residual to the final embedding or leaves it unchanged.

Residual adapter rather than full fine-tuning or fixed steering. Full fine-tuning can improve negated captions but risks disturbing the representation geometry used by general retrieval. Fixed activation steering avoids training the backbone, but the released configuration applies a shift to every text input and therefore changes affirmative captions as well Sammani et al. [2026]. NeRo-CLIP instead uses a small residual adapter that starts from the identity ($W_{\text{up}} = 0$) and is applied selectively through the regex router. The adapter is therefore not a replacement for CLIP’s text representation; it is a controlled residual correction used only in the routed regime.

Low-rank capacity. We set the adapter rank to $r = 8$, yielding 8,192 trainable parameters. This rank is intentionally small: the goal is not to learn a new text encoder, but a low-dimensional correction subspace for explicit negation. We use $r = 8$ as a conservative capacity margin while keeping the intervention more than three orders of magnitude smaller than the CLIP backbone.

4 Experiments

4.1 Setup

Backbone. NeRo-CLIP is built on CLIP ViT-B/32, using its frozen image and text encoders with embedding dimension 512.

Training data. The joint objective uses two NegBench Alhamoud et al. [2025] splits on MS-COCO val2017:

- **MCQ-Neg:** 5,914 4-way rows over 3,926 unique COCO images, with three caption templates in roughly equal proportion (positive 34.4%, hybrid 34.0%, negative 31.6%).
- **Retrieval-Neg:** 5,004 COCO val2017 images, 5 negated captions per image (25,000 pairs).

We use disjoint train/test partitions with seed 42 for both. MCQ-Neg uses a row-level 70/30 split (4,139 train, 1,775 test). Retrieval-Neg uses an image-level 80/20 split (4,000 train, 1,000 test) and the captions paired with them, giving $\sim 20,000$ training pairs and 5,004 test pairs.

Evaluation benchmarks. We evaluate five metrics:

- **MCQ-Neg.** The 1,775-row test split released with the NegBench reproducibility code. We use this row-level split for direct comparability with the NegBench release and the steering baseline.
- **Retrieval-Neg R@1.** 1,000-image test pool from our image-level 80/20 split, 5,004 caption-image pairs. The pool is disjoint from training images.
- **SimpleNeg Top-1.** 900 negated queries from Sammani et al. [2026] against a 25K-image pool. We use Qwen3-VL-8B as the automatic judge; the queries are free-form English. We validate the judge against human ratings on a 50-query subset (96% agreement).
- **N-COCO R@1.** A 10-query benchmark from the same source. We report it for direct comparison but interpret it cautiously because each correct retrieval changes R@1 by 0.10.
- **COCO val2017 retention R@1.** Standard text $\rightarrow$ image retrieval on the full 5,000-image val2017 set with 25,000 captions (5 per image; mostly affirmative). The regex router fires on 260/25,000 = 1.04% of these.

Baselines. We compare against CLIP without any correction, ConCLIP Singh et al. [2025], Neg-CLIP Yuksekgonul et al. [2023], CLIP-CC12M Alhamoud et al. [2025], and the layerwise activation steering baseline of Sammani et al. [2026]. We evaluate all baselines under the same protocol used for NeRo-CLIP: released checkpoints for ConCLIP, NegCLIP, and CC12M-NegFull from the NegBench release, and the steering baseline retrained from its published recipe at $\alpha=0.13$.

4.2 Main results

Table 1 reports NeRo-CLIP across 5 values of λ on the four negation metrics. Table 2 reports the retention metric. Figure 2 visualizes the accuracy–retention trade-off across methods; Figure 3 shows the per- λ curves for MCQ-Neg and Retrieval-Neg.

$\lambda=0.75$ trades a small MCQ loss for a large retrieval gain. The two single-objective configurations fail on opposite tasks. $\lambda=1$ (MCQ-only) overfits to within-row discrimination: MCQ-Neg 56.85, but Retrieval-Neg collapses to 25.68 (−18.98pp below CLIP). $\lambda=0$ (retrieval-only) overfits to image rank: Retrieval-Neg 49.06 (+4.40pp above CLIP), but MCQ-Neg drops to 37.80 (−1.86pp below CLIP). The joint mixture at $\lambda=0.75$ stays close to MCQ-only on MCQ-Neg (54.70 vs. 56.85, −2.15pp) while recovering Retrieval-Neg to 50.08 (+5.42pp above CLIP, +24.40pp vs. MCQ-only). Relative to CLIP, SimpleNeg gains +4.1pp (45.6 $\rightarrow$ 49.7) and N-COCO gains +0.10. The largest

Table 1: **NeRo-CLIP improves over CLIP on all four negation benchmarks.** All NeRo-CLIP rows use ViT-B/32 with a rank-8 adapter under per-caption routing, swept over the mixing coefficient λ (recommended $\lambda=0.75$ shaded); prior methods are re-measured under the same protocol. **Bold** marks the column maximum among baselines; underline marks the best NeRo-CLIP variant per column.

Method	MCQ-Neg ↑	Retrieval-Neg R@1 ↑	SimpleNeg Top-1 ↑	N-COCO R@1 ↑
<i>Baselines</i>				
CLIP	39.66	44.66	45.6	0.50
ConCLIP Singh et al. [2025]	25.07	44.86	31.8	0.70
NegCLIP Yuksekogonul et al. [2023]	26.40	60.77	47.8	0.20
CLIP-CC12M Alhamoud et al. [2025]	55.13	49.58	55.3	0.40
Layerwise steering Sammani et al. [2026]	41.60	36.17	52.6	0.80
<i>NeRo-CLIP (ours)</i>				
$\lambda=0.0$ (retrieval-only)	37.80	49.06	49.2	<u>0.60</u>
$\lambda=0.25$	42.08	<u>51.66</u>	48.9	<u>0.60</u>
$\lambda=0.5$	42.82	50.80	49.6	0.60
$\lambda=0.75$ (recommended)	54.70	50.08	<u>49.7</u>	<u>0.60</u>
$\lambda=1.0$ (MCQ-only)	<u>56.85</u>	25.68	44.4	0.50

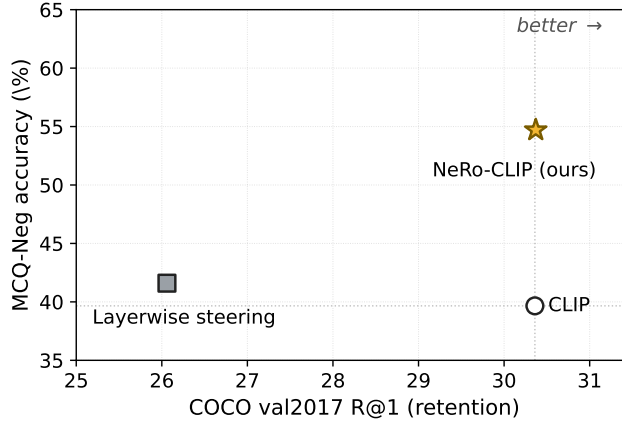


Figure 2: **NeRo-CLIP improves negation accuracy while preserving COCO retrieval.** Each point plots a method’s MCQ-Neg accuracy (y -axis) against its COCO val2017 R@1 retention (x -axis); the top-right corner is best. CLIP (open circle) sets the retention baseline with no correction. Layerwise steering buys MCQ-Neg accuracy by sacrificing retention, moving up but to the left (-4.30 pp on R@1). NeRo-CLIP primarily improves along the accuracy axis, gaining negation accuracy while holding retention within ± 0.05 pp of CLIP.

shift is on retrieval, where adding 25% retrieval supervision pulls the score from 18.98pp below CLIP to 5.42pp above it. When the loss only requires distinguishing four captions of the same image, the optimizer can shrink unrelated captions toward each other with no penalty; the retrieval term restores the cross-image alignment signal.

Comparison to layerwise steering. NeRo-CLIP and layerwise steering trade off differently across negation tasks. NeRo-CLIP is stronger on NegBench-style evaluations: MCQ-Neg improves from 41.60 to 54.70 (+13.10pp), and Retrieval-Neg R@1 from 36.17 to 50.08 (+13.91pp). Layerwise steering is stronger on open-pool retrieval: SimpleNeg reaches 52.6 vs. 49.7 (+2.9pp), and N-COCO reaches 0.80 vs. 0.60 (+0.20).

This split is consistent with the task format. NegBench-style evaluations use fixed candidates or fixed test pools; SimpleNeg and N-COCO require a query to pass strong image-level distractors in open-pool retrieval. Layerwise steering provides a larger text-side displacement, which helps with these Top-1 swaps. NeRo-CLIP provides a smaller, routed correction that is better aligned with the NegBench objectives.

Table 2: **Routing keeps general retrieval performance, unlike unconditional steering.** Text-to-image R@1 on COCO val2017 (5,000 images, 25,000 mostly affirmative captions); higher is better, and Δ is the change from CLIP. NeRo-CLIP is shown at its deployment setting ($\lambda=0.75$, per-caption routing)

Method	COCO R@1 $\uparrow$	Δ vs. CLIP
CLIP (baseline)	30.36	—
Layerwise steering Sammani et al. [2026]	26.06	−4.30
NeRo-CLIP (ours)	30.37	+0.01

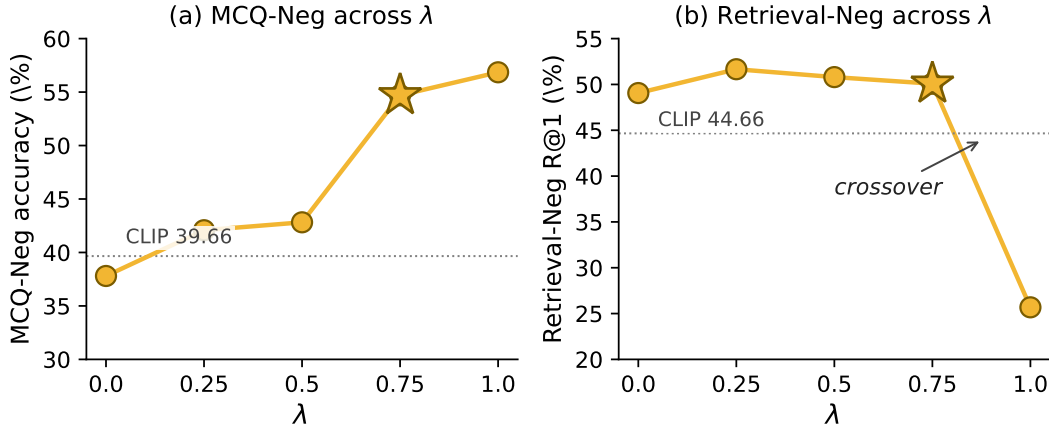


Figure 3: $\lambda=0.75$ is the point where MCQ discrimination and retrieval alignment are jointly satisfied. (a) MCQ-Neg accuracy across λ . (b) Retrieval-Neg R@1 across λ . Pushing λ to 1 (MCQ-only) collapses retrieval far below the CLIP baseline (25.68 vs. 44.66, dashed line), while every joint setting ($\lambda \leq 0.75$) sits +4 to +7pp above it. The two panels pull in opposite directions as $\lambda \rightarrow 1$, and $\lambda=0.75$ is the largest value that keeps MCQ strong without giving up retrieval.

The two methods therefore expose different accuracy–retention operating points; we analyze the retention side of this trade-off in Section 4.3.

4.3 Retention: COCO val2017 retrieval

Table 2 reports text-to-image retrieval R@1 on the full COCO val2017 split. NeRo-CLIP scores 30.37, matching the CLIP baseline of 30.36. This near-perfect retention is consistent with the low routing rate: on COCO val2017, the regex matches only 260/25,000 captions (1.04%), so the remaining 98.96% keep their original CLIP text embedding.

Layerwise steering scores 26.06 under the same protocol, a −4.30pp retention cost. The released configuration applies activation shifts to every text input across all text-encoder layers. NeRo-CLIP instead changes only the small subset of captions selected by the router, which explains why its retention remains at the CLIP baseline.

4.4 Mechanism: why $\lambda=0.75$

Figure 3a shows MCQ-Neg accuracy across λ . The adapter trained with MCQ-only supervision ($\lambda=1$) reaches higher MCQ-Neg in isolation, but the same setting damages general retrieval. Our deployment configuration uses $\lambda=0.75$, which reaches 54.70 on MCQ-Neg while preserving COCO retention.

Figure 3b shows Retrieval-Neg R@1 as a function of λ . At $\lambda=1$, the adapter loses 18.98pp relative to CLIP; at $\lambda=0$, it gains 4.40pp. All joint configurations with $\lambda \leq 0.75$ land above the CLIP baseline. MCQ supervision alone leaves cross-image alignment unconstrained, while retrieval supervision alone gives no caption-vs-caption signal. The mixture at $\lambda=0.75$ keeps MCQ discrimination strong (+15.04pp over CLIP) without the retrieval cost.

5 Limitations

NeRo-CLIP shows that a lightweight routed adapter can improve explicit negation without retraining CLIP, but several aspects of the setting remain underexplored. The main open questions concern backbone coverage, router coverage, and transfer to retrieval-heavy negation settings.

Backbone coverage. We use CLIP ViT-B/32 as the reference backbone because it offers a controlled middle ground. It is large enough for the retrieval setting to be nontrivial, yet small enough to make repeated sweeps over routing, λ , and adapter variants feasible. Fixing this backbone also separates the question we study from backbone scaling: the experiment tests whether a small text-side correction can change negation behavior when the underlying CLIP encoder is held constant. The same choice also makes comparisons to existing B/32-based negation correction and steering methods easier to interpret. This comparability is useful, but the evidence remains tied to this backbone choice. Whether the same trade-off holds for ViT-B/16, ViT-L/14, or newer CLIP-style checkpoints remains open. Each setting would require a new cache, a newly trained adapter, and a separate λ sweep.

Router coverage. The regex router is intentionally conservative. It preserves ordinary COCO retrieval by leaving most affirmative captions untouched, which is central to NeRo-CLIP’s mixed-traffic setting. This choice narrows coverage: the current rule set mainly targets explicit markers such as “no”, “not”, and “without”. Broader negation forms would require either a larger dictionary or a learned router. A learned router is a natural extension, but its objective should remain asymmetric. Recall on negated captions matters, but false positives on affirmative captions matter just as much. If the router matches too broadly, the adapter approaches an unconditional intervention, removing the retention benefit that routing is meant to provide.

Transfer to retrieval-heavy settings. NeRo-CLIP is designed for a specific regime: explicit negation in a frozen CLIP retrieval pipeline, with a small adapter applied only when the router detects the negation markers. This design keeps the correction parameter-efficient and easy to gate, while placing the method in a mixed-traffic retrieval regime. Broader negation forms and retrieval-heavy settings may require richer routing, stronger interventions, or additional training signals. NeRo-CLIP is therefore best read as a parameter-efficient post-hoc correction for explicit negation under routed deployment. Extending the same approach beyond this setting would require broader routing coverage and task-specific evaluation.

6 Conclusion

We study whether explicit negation failures in CLIP can be corrected without retraining the backbone. NeRo-CLIP addresses this question with a small text-side residual adapter, a router, and a joint MCQ–retrieval training objective. The method keeps the CLIP encoders fixed and applies the correction only when the input contains an explicit negation marker. The experiments show that this design improves explicit negation while preserving ordinary retrieval behavior. The routing rule keeps most affirmative COCO captions unchanged, which prevents the adapter from becoming a uniform shift over all inputs. The joint objective also matters: MCQ-only training improves row-level discrimination, but retrieval supervision is needed to keep the adapted captions calibrated against the image pool. NeRo-CLIP is therefore best understood as a parameter-efficient post-hoc correction for a specific deployment regime: mixed retrieval traffic with explicit negation. It does not require changing CLIP’s backbone or pretraining data, and its main value lies in the combination of small intervention size, routed application, and retrieval-aware training.

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